KERALA POLYTECHNIC · REVISION 2026

Electric and Hybrid Vehicles

Complete student lesson for course code **6051A**.

Revision 2026 Semester 6 Program Elective 4 modules

Course snapshot

Scheme: 4 (L:4, T:0, P:0)

Credits: 4

Contact: 60 periods per semester

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: The supplied PDF does not specify a separate CIE/ESE distribution. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Handbook method: Do not treat a topic as completed after reading its heading. For every topic card, complete the learning objectives, follow the conceptual framework, write the worked answer method, and answer the self-check questions. This creates a study record that is useful for class learning and examination preparation.

Course Dashboard

Course code

6051A

Semester

6

Credits

4

Teaching scheme

4 (L:4, T:0, P:0)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Electric and Hybrid Vehicle Fundamentals	12	CO1
2	EV Design and Motor Drives	16	CO2

No.	Module	Official hours	Relevant CO / level
3	Power Electronics, Drive Trains and Braking	14	CO3
4	Energy Storage, Fuel Cells and Charging	16	CO4

Prerequisites

- Knowledge on basic Automobile Engineering — Basic Automobile Engineering, Semester 2.
- Knowledge on Automobile electrical and electronics systems — Automobile Electrical and Electronic systems, Semester 4.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. To understand general aspects of Electric and Hybrid Vehicles.
2. To understand basic design considerations of EVs.
3. To select the suitable electric propulsion systems.
4. To select required energy storage and charging devices.

Course Outcomes

1. Summarize the general aspects of Electric and Hybrid Vehicles.
2. Outline the basic design considerations of Electric vehicles.
3. Identify various drive trains in Electric vehicles and Hybrid electric vehicles.

4. Select the required energy storage and charging devices for Electric and Hybrid vehicles.

CO–PO mapping as supplied

The supplied CO–PO table is reproduced at outcome level from the PDF; blank cells are not treated as mapped.
Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Official syllabus point-by-point coverage

This audit layer maps every official source block to the corresponding lesson module. The source transcript is retained verbatim as extracted from the supplied syllabus; the study guidance below is educational support and is not presented as additional official syllabus content. No official point is intentionally omitted.

Source-to-lesson coverage map

Module	Lesson topic cards	Source basis	Official point units
Module 1	6	Official Contents block	13
Module 2	4	Official Contents block	6
Module 3	6	Official Contents block	15
Module 4	5	Official Contents block	5

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Need for hybrid and electric vehicles	1
module-1	M1.02	Components and working principle of electric and hybrid electric vehicles	3
module-1	M1.03	Configurations of electric vehicles and hybrid vehicles	4

Module	Topic code	Topic	Hours
module-1	M1.04	Compare petrol, diesel, hybrid and electric vehicles	2
module-1	M1.05	Advantages and disadvantages of hybrid and electric vehicles	1
module-1	M1.06	Government policies and schemes related to Electric vehicles	1
module-2	M2.01	Design requirements for electric vehicles	3
module-2	M2.02	Performance during acceleration, coasting, moving up and down a hill	5
module-2	M2.03	Types of motors used in electric vehicles	3
module-2	M2.04	Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives	5
module-3	M3.01	Power electronics converters used in electric and hybrid vehicles	2
module-3	M3.02	Control systems for EV	2
module-3	M3.03	Drive trains in Electric and Hybrid Electric Vehicles	2
module-3	M3.04	Drive train configurations in electric and hybrid electric vehicles	3
module-3	M3.05	Regenerative braking and hybrid braking	2
module-3	M3.06	Service and maintenance of EV components	3
module-4	M4.01	Energy storage requirements in electric and hybrid vehicles	2
module-4	M4.02	Construction and working of batteries used in electric and hybrid vehicles	5
module-4	M4.03	Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages	3
module-4	M4.04	Operation of Fuel cells	2

Module	Topic code	Topic	Hours
module-4	M4.05	Battery charging Technologies and Battery management systems	4

Learning Roadmap

1. Electric and Hybrid Vehicle Fundamentals
CO1 · 12 hours

2. EV Design and Motor Drives
CO2 · 16 hours

3. Power Electronics, Drive Trains and Braking
CO3 · 14 hours

4. Energy Storage, Fuel Cells and Charging
CO4 · 16 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Electric and Hybrid Vehicle Fundamentals

This module introduces why electric and hybrid propulsion is required, the main vehicle configurations, comparative performance, and the policy context.

Hours: 12
CO1 · Bloom level: Understanding

Handbook study routine: Complete Module 1 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 1

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.
Main components and working principles of electric and hybrid vehicles.
Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV)
– Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential.
Types of HEV - series, parallel, series-parallel and Complex.
Comparative study of EV and conventional vehicles, Differences between complete EV and
Hybrid vehicles. Life cycle assessment of electric vehicles.
Government Schemes and Progress: FAME-1, FAME-2; Transformative -mobility and

Energy Storage Mission, and other latest central level policies - State Policies, subsidies, and incentives - Global experiences and success stories on EV production.

Point-by-point study guide

Official syllabus point 1: Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.

Exact source point

Syllabus text: Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.

How to understand and study it

This point is an official syllabus requirement: “Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Need for hybrid, electric vehicles, Historical background. Social, environmental importance of hybrid □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. using the official terminology retained above.
- Organise the important elements of Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. into a logical sequence, classification, structure, or calculation.
- Connect Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles.?
- How would you distinguish Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. from a closely related idea?
- Where would an engineer or technician encounter Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles., and what must be checked before using it?
- What common mistake would make an answer or operation involving Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. □□□□ topic □□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□□□.

Official syllabus point 2: Main components and working principles of electric and hybrid vehicles.

Exact source point

Syllabus text: Main components and working principles of electric and hybrid vehicles.

How to understand and study it

This point is an official syllabus requirement: “Main components and working principles of electric and hybrid vehicles.”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ Main components, working principles of electric, hybrid vehicles. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Main components and working principles of electric and hybrid vehicles. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Main components and working principles of electric and hybrid vehicles. using the official terminology retained above.
- Organise the important elements of Main components and working principles of electric and hybrid vehicles. into a logical sequence, classification, structure, or calculation.
- Connect Main components and working principles of electric and hybrid vehicles. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Main components and working principles of electric and hybrid vehicles. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Main components and working principles of electric and hybrid vehicles.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Main components and working principles of electric and hybrid vehicles. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Main components and working principles of electric and hybrid vehicles., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Main components and working principles of electric and hybrid vehicles., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Main components and working principles of electric and hybrid vehicles.?
- How would you distinguish Main components and working principles of electric and hybrid vehicles. from a closely related idea?
- Where would an engineer or technician encounter Main components and working principles of electric and hybrid vehicles., and what must be checked before using it?
- What common mistake would make an answer or operation involving Main components and working principles of electric and hybrid vehicles. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Main components and working principles of electric and hybrid vehicles. [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 3: Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery

Exact source point

Syllabus text: Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery

How to understand and study it

This point is an official syllabus requirement: “Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery using the official terminology retained above.
- Organise the important elements of Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery into a logical sequence, classification, structure, or calculation.
- Connect Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery?
- How would you distinguish Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery from a closely related idea?
- Where would an engineer or technician encounter Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery, and what must be checked before using it?
- What common mistake would make an answer or operation involving Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 4: Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV)

Exact source point

Syllabus text: Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV)

How to understand and study it

This point is an official syllabus requirement: “Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV)”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) using the official terminology retained above.
- Organise the important elements of Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) into a logical sequence, classification, structure, or calculation.
- Connect Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV)?
- How would you distinguish Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) from a closely related idea?
- Where would an engineer or technician encounter Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV), and what must be checked before using it?
- What common mistake would make an answer or operation involving Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 5: – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential.

Exact source point

Syllabus text: – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential.

How to understand and study it

This point is an official syllabus requirement: "– Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential.". Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with, without differential. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

– Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. using the official terminology retained above.
- Organise the important elements of – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. into a logical sequence, classification, structure, or calculation.
- Connect – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential.?
- How would you distinguish – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. from a closely related idea?
- Where would an engineer or technician encounter – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential., and what must be checked before using it?
- What common mistake would make an answer or operation involving – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.

- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and without differential. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 6: Types of HEV - series, parallel, series-parallel and Complex.

Exact source point

Syllabus text: Types of HEV - series, parallel, series-parallel and Complex.

How to understand and study it

This point is an official syllabus requirement: “Types of HEV - series, parallel, series-parallel and Complex.”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Types of HEV - series, parallel, series-parallel, Complex. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Types of HEV - series, parallel, series-parallel and Complex. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Types of HEV - series, parallel, series-parallel and Complex. using the official terminology retained above.
- Organise the important elements of Types of HEV - series, parallel, series-parallel and Complex. into a logical sequence, classification, structure, or calculation.
- Connect Types of HEV - series, parallel, series-parallel and Complex. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Types of HEV - series, parallel, series-parallel and Complex. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Types of HEV - series, parallel, series-parallel and Complex.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Types of HEV - series, parallel, series-parallel and Complex. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Types of HEV - series, parallel, series-parallel and Complex., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Types of HEV - series, parallel, series-parallel and Complex., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Types of HEV - series, parallel, series-parallel and Complex.?
- How would you distinguish Types of HEV - series, parallel, series-parallel and Complex. from a closely related idea?
- Where would an engineer or technician encounter Types of HEV - series, parallel, series-parallel and Complex., and what must be checked before using it?
- What common mistake would make an answer or operation involving Types of HEV - series, parallel, series-parallel and Complex. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Types of HEV - series, parallel, series-parallel and Complex. [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 7: Comparative study of EV and conventional vehicles, Differences between complete EV and

Exact source point

> **Syllabus text:** Comparative study of EV and conventional vehicles, Differences between complete EV and

How to understand and study it

This point is an official syllabus requirement: “Comparative study of EV and conventional vehicles, Differences between complete EV and”. Prepare a comparison using the same criteria for every item: principle, performance, cost or complexity, limitation, and application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** ▯ syllabus point ▯▯▯▯▯▯▯▯▯▯▯ Comparative study of EV, conventional vehicles, Differences between complete EV and ▯▯▯▯ technical terms English-▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯. ▯▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯ principle ▯▯▯▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯ ▯▯▯ term-▯▯▯▯ function, difference, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Diagram, sequence, formula, unit ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ revision ▯▯▯▯▯▯▯.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Comparative study of EV and conventional vehicles, Differences between complete EV and is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Comparative study of EV and conventional vehicles, Differences between complete EV and using the official terminology retained above.
- Organise the important elements of Comparative study of EV and conventional vehicles, Differences between complete EV and into a logical sequence, classification, structure, or calculation.
- Connect Comparative study of EV and conventional vehicles, Differences between complete EV and with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Comparative study of EV and conventional vehicles, Differences between complete EV and with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Comparative study of EV and conventional vehicles, Differences between complete EV and. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Comparative study of EV and conventional vehicles, Differences between complete EV and is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Comparative study of EV and conventional vehicles, Differences between complete EV and, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Comparative study of EV and conventional vehicles, Differences between complete EV and, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Comparative study of EV and conventional vehicles, Differences between complete EV and?
- How would you distinguish Comparative study of EV and conventional vehicles, Differences between complete EV and from a closely related idea?
- Where would an engineer or technician encounter Comparative study of EV and conventional vehicles, Differences between complete EV and, and what must be checked before using it?
- What common mistake would make an answer or operation involving Comparative study of EV and conventional vehicles, Differences between complete EV and incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Comparative study of EV and conventional vehicles, Differences between complete EV and [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 8: Hybrid vehicles. Life cycle assessment of electric vehicles.

Exact source point

Syllabus text: Hybrid vehicles. Life cycle assessment of electric vehicles.

How to understand and study it

This point is an official syllabus requirement: “Hybrid vehicles. Life cycle assessment of electric vehicles.”. Study the sequence from input or initial condition to operation and final output. Note the role of each named stage and the cause-and-effect relationship. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ Hybrid vehicles. Life cycle assessment of electric vehicles. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Hybrid vehicles. Life cycle assessment of electric vehicles. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Hybrid vehicles. Life cycle assessment of electric vehicles. using the official terminology retained above.
- Organise the important elements of Hybrid vehicles. Life cycle assessment of electric vehicles. into a logical sequence, classification, structure, or calculation.
- Connect Hybrid vehicles. Life cycle assessment of electric vehicles. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Hybrid vehicles. Life cycle assessment of electric vehicles. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Hybrid vehicles. Life cycle assessment of electric vehicles.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Hybrid vehicles. Life cycle assessment of electric vehicles. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Hybrid vehicles. Life cycle assessment of electric vehicles., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Hybrid vehicles. Life cycle assessment of electric vehicles., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Hybrid vehicles. Life cycle assessment of electric vehicles.?
- How would you distinguish Hybrid vehicles. Life cycle assessment of electric vehicles. from a closely related idea?
- Where would an engineer or technician encounter Hybrid vehicles. Life cycle assessment of electric vehicles., and what must be checked before using it?
- What common mistake would make an answer or operation involving Hybrid vehicles. Life cycle assessment of electric vehicles. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Hybrid vehicles. Life cycle assessment of electric vehicles. [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 9: Government Schemes and Progress: FAME-1, FAME-2

Exact source point

Syllabus text: Government Schemes and Progress: FAME-1, FAME-2

How to understand and study it

This point is an official syllabus requirement: “Government Schemes and Progress: FAME-1, FAME-2”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Government Schemes, Progress, FAME-1, FAME-2 □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Government Schemes and Progress: FAME-1, FAME-2 is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Government Schemes and Progress: FAME-1, FAME-2 using the official terminology retained above.
- Organise the important elements of Government Schemes and Progress: FAME-1, FAME-2 into a logical sequence, classification, structure, or calculation.
- Connect Government Schemes and Progress: FAME-1, FAME-2 with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Government Schemes and Progress: FAME-1, FAME-2 with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Government Schemes and Progress: FAME-1, FAME-2. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Government Schemes and Progress: FAME-1, FAME-2 is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Government Schemes and Progress: FAME-1, FAME-2, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Government Schemes and Progress: FAME-1, FAME-2, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Government Schemes and Progress: FAME-1, FAME-2?
- How would you distinguish Government Schemes and Progress: FAME-1, FAME-2 from a closely related idea?
- Where would an engineer or technician encounter Government Schemes and Progress: FAME-1, FAME-2, and what must be checked before using it?
- What common mistake would make an answer or operation involving Government Schemes and Progress: FAME-1, FAME-2 incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Government Schemes and Progress: FAME-1, FAME-2 [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 10: Transformative -mobility and

Exact source point

Syllabus text: Transformative -mobility and

How to understand and study it

This point is an official syllabus requirement: “Transformative -mobility and”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Transformative -mobility and □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□□ □□□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Transformative -mobility and is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Transformative -mobility and using the official terminology retained above.
- Organise the important elements of Transformative -mobility and into a logical sequence, classification, structure, or calculation.
- Connect Transformative -mobility and with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Transformative -mobility and with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Transformative -mobility and. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Transformative -mobility and is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Transformative -mobility and, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Transformative -mobility and, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Transformative -mobility and?
- How would you distinguish Transformative -mobility and from a closely related idea?
- Where would an engineer or technician encounter Transformative -mobility and, and what must be checked before using it?
- What common mistake would make an answer or operation involving Transformative -mobility and incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Transformative -mobility and □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 11: Energy Storage Mission, and other latest central level policies

Exact source point

Syllabus text: Energy Storage Mission, and other latest central level policies

How to understand and study it

This point is an official syllabus requirement: “Energy Storage Mission, and other latest central level policies”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Energy Storage Mission, and other latest central level policies □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Energy Storage Mission, and other latest central level policies must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Energy Storage Mission, and other latest central level policies using the official terminology retained above.
- Organise the important elements of Energy Storage Mission, and other latest central level policies into a logical sequence, classification, structure, or calculation.
- Connect Energy Storage Mission, and other latest central level policies with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Energy Storage Mission, and other latest central level policies with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Energy Storage Mission, and other latest central level policies. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Energy Storage Mission, and other latest central level policies is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Energy Storage Mission, and other latest central level policies, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Energy Storage Mission, and other latest central level policies, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Energy Storage Mission, and other latest central level policies?
- How would you distinguish Energy Storage Mission, and other latest central level policies from a closely related idea?
- Where would an engineer or technician encounter Energy Storage Mission, and other latest central level policies, and what must be checked before using it?
- What common mistake would make an answer or operation involving Energy Storage Mission, and other latest central level policies incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Energy Storage Mission, and other latest central level policies [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 12: State Policies, subsidies, and incentives

Exact source point

Syllabus text: State Policies, subsidies, and incentives

How to understand and study it

This point is an official syllabus requirement: “State Policies, subsidies, and incentives”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□□ State Policies, subsidies, and incentives □□□□ technical terms English-□□□□□□□□ □□□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

State Policies, subsidies, and incentives is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope State Policies, subsidies, and incentives using the official terminology retained above.
- Organise the important elements of State Policies, subsidies, and incentives into a logical sequence, classification, structure, or calculation.
- Connect State Policies, subsidies, and incentives with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on State Policies, subsidies, and incentives with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading State Policies, subsidies, and incentives. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of State Policies, subsidies, and incentives is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on State Policies, subsidies, and incentives, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is State Policies, subsidies, and incentives, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining State Policies, subsidies, and incentives?
- How would you distinguish State Policies, subsidies, and incentives from a closely related idea?
- Where would an engineer or technician encounter State Policies, subsidies, and incentives, and what must be checked before using it?
- What common mistake would make an answer or operation involving State Policies, subsidies, and incentives incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: State Policies, subsidies, and incentives [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 13: Global experiences and success stories on EV production.

Exact source point

> **Syllabus text:** Global experiences and success stories on EV production.

How to understand and study it

This point is an official syllabus requirement: “Global experiences and success stories on EV production.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Global experiences, success stories on EV production. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Global experiences and success stories on EV production. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Global experiences and success stories on EV production. using the official terminology retained above.
- Organise the important elements of Global experiences and success stories on EV production. into a logical sequence, classification, structure, or calculation.
- Connect Global experiences and success stories on EV production. with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on Global experiences and success stories on EV production. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Global experiences and success stories on EV production.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Global experiences and success stories on EV production. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Global experiences and success stories on EV production., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Global experiences and success stories on EV production., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Global experiences and success stories on EV production.?
- How would you distinguish Global experiences and success stories on EV production. from a closely related idea?
- Where would an engineer or technician encounter Global experiences and success stories on EV production., and what must be checked before using it?
- What common mistake would make an answer or operation involving Global experiences and success stories on EV production. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Global experiences and success stories on EV production. □□□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

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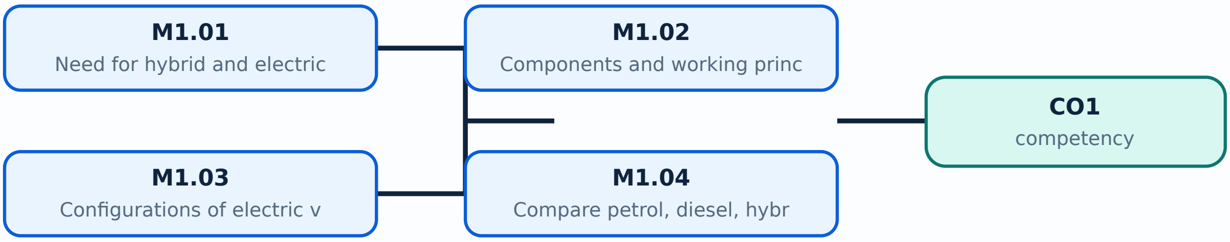
Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

M1.01 — Need for hybrid and electric vehicles (1 hours)

Concept and explanation

<p>Scope. The official syllabus places Need for hybrid and electric vehicles in this topic. The required scope is: Need for hybrid and electric vehicles; historical background; social and environmental importance.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Need for hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Need for hybrid and electric vehicles □□□□ topic □□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Need for hybrid and electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Need for hybrid and electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Need for hybrid and electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Need for hybrid and electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M1.01 — Need for hybrid and electric vehicles (1 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.01 — Need for hybrid and electric vehicles (1 hours) using the official terminology retained above.
- Organise the important elements of M1.01 — Need for hybrid and electric vehicles (1 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.01 — Need for hybrid and electric vehicles (1 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.01 — Need for hybrid and electric vehicles (1 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.01 — Need for hybrid and electric vehicles (1 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.01 — Need for hybrid and electric vehicles (1 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.01 — Need for hybrid and electric vehicles (1 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.01 — Need for hybrid and electric vehicles (1 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.01 — Need for hybrid and electric vehicles (1 hours)?
- How would you distinguish M1.01 — Need for hybrid and electric vehicles (1 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.01 — Need for hybrid and electric vehicles (1 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.01 — Need for hybrid and electric vehicles (1 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.01 — Need for hybrid and electric vehicles (1 hours)
[illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Components and working principle of electric and hybrid electric vehicles in this topic. The required scope is: Main components and working principles of electric and hybrid vehicles.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Components and working principle of electric and hybrid electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Components and working principle of electric and hybrid electric vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Components and working principle of electric and hybrid electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Components and working principle of electric and hybrid electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Components and working principle of electric and hybrid electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Components and working principle of electric and hybrid electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) using the official terminology retained above.
- Organise the important elements of M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours)?
- How would you distinguish M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.02 — Components and working principle of electric and hybrid electric vehicles (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours)

Concept and explanation

<p>Scope. The official syllabus places Configurations of electric vehicles and hybrid vehicles in this topic. The required scope is: Pure Electric vehicle (PE), Battery Electric vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric vehicle (HEV), Plug-in Hybrid Electric vehicle (PHEV), and types based on differential; series, parallel, series-parallel and complex HEV.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"> <table><caption>Study frame for Configurations of electric vehicles and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th> <td>How the input is converted, controlled, transported, stored, measured, or protected. </td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr> <th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p> Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Configurations of electric vehicles and hybrid vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Configurations of electric vehicles and hybrid vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Configurations of electric vehicles and hybrid vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Configurations of electric vehicles and hybrid vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Configurations of electric vehicles and hybrid vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) using the official terminology retained above.
- Organise the important elements of M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours)?
- How would you distinguish M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.03 — Configurations of electric vehicles and hybrid vehicles (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Compare petrol, diesel, hybrid and electric vehicles in this topic. The required scope is: Comparative study of EV and conventional vehicles; differences between complete EV and hybrid vehicles; life cycle assessment.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Compare petrol, diesel, hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Malayalam support: Malayalam support: Compare petrol, diesel, hybrid and electric vehicles □□□□ topic □□□□□□□□□□□□ □□□□□□ □□□□□□□□ purpose, working principle, □□□□□□□ components □□□□□□□□□□□ variables, □□□□□□□ performance □□□□□□□□□□□ safety checks □□□□ □□□□□□□□□□□ □□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□; diagram □□□□□□□□□□□□ step sequence □□□□□□□□□□□ process □□□□□□□□□□□□ □□□□□□□□□□□□□□□□.

Study points

- Define Compare petrol, diesel, hybrid and electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Compare petrol, diesel, hybrid and electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Compare petrol, diesel, hybrid and electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Compare petrol, diesel, hybrid and electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) using the official terminology retained above.
- Organise the important elements of M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours)?
- How would you distinguish M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.04 — Compare petrol, diesel, hybrid and electric vehicles (2 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours)

Concept and explanation

<p>Scope. The official syllabus places Advantages and disadvantages of hybrid and electric vehicles in this topic. The required scope is: Technical, economic, environmental, range, charging, maintenance, and lifecycle advantages and limitations.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Advantages and disadvantages of hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Advantages and disadvantages of hybrid and electric vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Advantages and disadvantages of hybrid and electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Advantages and disadvantages of hybrid and electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Advantages and disadvantages of hybrid and electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Advantages and disadvantages of hybrid and electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) using the official terminology retained above.
- Organise the important elements of M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours)?
- How would you distinguish M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.05 — Advantages and disadvantages of hybrid and electric vehicles (1 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

M1.06 — Government policies and schemes related to Electric vehicles (1 hours)

Concept and explanation

<p>Scope. The official syllabus places Government policies and schemes related to Electric vehicles in this topic. The required scope is: FAME-1, FAME-2, Transformative Mobility and Energy Storage Mission, central and state policies, subsidies, incentives, and global experiences.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Government policies and schemes related to Electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Government policies and schemes related to Electric vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Government policies and schemes related to Electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.

- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Government policies and schemes related to Electric vehicles is applied in electric vehicle engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Government policies and schemes related to Electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Government policies and schemes related to Electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M1.06 — Government policies and schemes related to Electric vehicles (1 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.06 — Government policies and schemes related to Electric vehicles (1 hours) using the official terminology retained above.
- Organise the important elements of M1.06 — Government policies and schemes related to Electric vehicles (1 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.06 — Government policies and schemes related to Electric vehicles (1 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Electric and Hybrid Vehicle Fundamentals.
- Write an examination answer on M1.06 — Government policies and schemes related to Electric vehicles (1 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.06 — Government policies and schemes related to Electric vehicles (1 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.06 — Government policies and schemes related to Electric vehicles (1 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.06 — Government policies and schemes related to Electric vehicles (1 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.06 — Government policies and schemes related to Electric vehicles (1 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.06 — Government policies and schemes related to Electric vehicles (1 hours)?
- How would you distinguish M1.06 — Government policies and schemes related to Electric vehicles (1 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.06 — Government policies and schemes related to Electric vehicles (1 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.06 — Government policies and schemes related to Electric vehicles (1 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.06 — Government policies and schemes related to Electric vehicles (1 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Module summary: Electric vehicle architecture and policy comparison.

OFFICIAL MODULE 2

EV Design and Motor Drives

Design begins with the required range, speed, acceleration, gradeability, mass, tractive effort, transmission efficiency, and energy consumption.

Hours: 16
CO2 · Bloom level: Understanding

Handbook study routine: Complete Module 2 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 2

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

:
Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission
Requirement, Transmission efficiency, energy consumption.

Motor drive Technologies: Electric propulsion unit - Classification of EV Motors, Configuration and control of DC Motor drives, Induction Motor drives, Permanent Magnet Motor drives, Switched reluctance motor dives.

Identify various drive trains in Electric Vehicles and Hybrid Electric

Point-by-point study guide

Official syllabus point 1: Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission

Exact source point

> **Syllabus text:** Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission

How to understand and study it

This point is an official syllabus requirement: “Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission using the official terminology retained above.
- Organise the important elements of Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission into a logical sequence, classification, structure, or calculation.
- Connect Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.

- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission, and what is its scope in this module?

- Which elements, stages, types, quantities, or conditions must be named when explaining Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission?
- How would you distinguish Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission from a closely related idea?
- Where would an engineer or technician encounter Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission, and what must be checked before using it?
- What common mistake would make an answer or operation involving Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Design requirement for electric vehicles- Range, maximum velocity, acceleration, power requirement, mass of the vehicle. Various Resistances, Performance of electric vehicles-acceleration, coasting, moving up and down a hill. Tractive Effort and Transmission □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 2: Requirement, Transmission efficiency, energy consumption.

Exact source point

Syllabus text: Requirement, Transmission efficiency, energy consumption.

How to understand and study it

This point is an official syllabus requirement: “Requirement, Transmission efficiency, energy consumption.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Requirement, Transmission efficiency, energy consumption. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Requirement, Transmission efficiency, energy consumption. must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Requirement, Transmission efficiency, energy consumption. using the official terminology retained above.
- Organise the important elements of Requirement, Transmission efficiency, energy consumption. into a logical sequence, classification, structure, or calculation.
- Connect Requirement, Transmission efficiency, energy consumption. with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on Requirement, Transmission efficiency, energy consumption. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Requirement, Transmission efficiency, energy consumption.. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Requirement, Transmission efficiency, energy consumption. is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Requirement, Transmission efficiency, energy consumption., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Requirement, Transmission efficiency, energy consumption., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Requirement, Transmission efficiency, energy consumption.?
- How would you distinguish Requirement, Transmission efficiency, energy consumption. from a closely related idea?
- Where would an engineer or technician encounter Requirement, Transmission efficiency, energy consumption., and what must be checked before using it?
- What common mistake would make an answer or operation involving Requirement, Transmission efficiency, energy consumption. incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Requirement, Transmission efficiency, energy consumption. [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 3: Motor drive Technologies: Electric propulsion unit - Classification of EV Motors

Exact source point

> **Syllabus text:** Motor drive Technologies: Electric propulsion unit - Classification of EV Motors

How to understand and study it

This point is an official syllabus requirement: “Motor drive Technologies: Electric propulsion unit - Classification of EV Motors”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** ▯ syllabus point ▯▯▯▯▯▯▯▯▯▯▯▯ Motor drive Technologies, Electric propulsion unit - Classification of EV Motors ▯▯▯▯ technical terms English-▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯. ▯▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯ principle ▯▯▯▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯ ▯▯▯ term-▯▯▯▯ function, difference, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Diagram, sequence, formula, unit ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ revision ▯▯▯▯▯▯▯.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Motor drive Technologies: Electric propulsion unit - Classification of EV Motors must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Motor drive Technologies: Electric propulsion unit - Classification of EV Motors using the official terminology retained above.
- Organise the important elements of Motor drive Technologies: Electric propulsion unit - Classification of EV Motors into a logical sequence, classification, structure, or calculation.
- Connect Motor drive Technologies: Electric propulsion unit - Classification of EV Motors with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on Motor drive Technologies: Electric propulsion unit - Classification of EV Motors with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Motor drive Technologies: Electric propulsion unit - Classification of EV Motors. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Motor drive Technologies: Electric propulsion unit - Classification of EV Motors is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Motor drive Technologies: Electric propulsion unit - Classification of EV Motors, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Motor drive Technologies: Electric propulsion unit - Classification of EV Motors, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Motor drive Technologies: Electric propulsion unit - Classification of EV Motors?
- How would you distinguish Motor drive Technologies: Electric propulsion unit - Classification of EV Motors from a closely related idea?
- Where would an engineer or technician encounter Motor drive Technologies: Electric propulsion unit - Classification of EV Motors, and what must be checked before using it?
- What common mistake would make an answer or operation involving Motor drive Technologies: Electric propulsion unit - Classification of EV Motors incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Motor drive Technologies: Electric propulsion unit - Classification of EV Motors [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 4: Configuration and control of DC Motor drives, Induction Motor drives, Permanent

Exact source point

Syllabus text: Configuration and control of DC Motor drives, Induction Motor drives, Permanent

How to understand and study it

This point is an official syllabus requirement: “Configuration and control of DC Motor drives, Induction Motor drives, Permanent”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Configuration, control of DC Motor drives, Induction Motor drives, Permanent □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Configuration and control of DC Motor drives, Induction Motor drives, Permanent is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope Configuration and control of DC Motor drives, Induction Motor drives, Permanent using the official terminology retained above.
- Organise the important elements of Configuration and control of DC Motor drives, Induction Motor drives, Permanent into a logical sequence, classification, structure, or calculation.
- Connect Configuration and control of DC Motor drives, Induction Motor drives, Permanent with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on Configuration and control of DC Motor drives, Induction Motor drives, Permanent with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Configuration and control of DC Motor drives, Induction Motor drives, Permanent. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, Configuration and control of DC Motor drives, Induction Motor drives, Permanent is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on Configuration and control of DC Motor drives, Induction Motor drives, Permanent, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Configuration and control of DC Motor drives, Induction Motor drives, Permanent, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Configuration and control of DC Motor drives, Induction Motor drives, Permanent?
- How would you distinguish Configuration and control of DC Motor drives, Induction Motor drives, Permanent from a closely related idea?
- Where would an engineer or technician encounter Configuration and control of DC Motor drives, Induction Motor drives, Permanent, and what must be checked before using it?
- What common mistake would make an answer or operation involving Configuration and control of DC Motor drives, Induction Motor drives, Permanent incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: Configuration and control of DC Motor drives, Induction Motor drives, Permanent [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 5: Magnet Motor drives, Switched reluctance motor dives.

Exact source point

Syllabus text: Magnet Motor drives, Switched reluctance motor dives.

How to understand and study it

This point is an official syllabus requirement: “Magnet Motor drives, Switched reluctance motor dives.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Magnet Motor drives, Switched reluctance motor dives. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Magnet Motor drives, Switched reluctance motor dives. is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope Magnet Motor drives, Switched reluctance motor dives. using the official terminology retained above.
- Organise the important elements of Magnet Motor drives, Switched reluctance motor dives. into a logical sequence, classification, structure, or calculation.
- Connect Magnet Motor drives, Switched reluctance motor dives. with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on Magnet Motor drives, Switched reluctance motor dives. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Magnet Motor drives, Switched reluctance motor dives.. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, Magnet Motor drives, Switched reluctance motor dives. is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on Magnet Motor drives, Switched reluctance motor dives., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Magnet Motor drives, Switched reluctance motor dives., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Magnet Motor drives, Switched reluctance motor dives.?
- How would you distinguish Magnet Motor drives, Switched reluctance motor dives. from a closely related idea?
- Where would an engineer or technician encounter Magnet Motor drives, Switched reluctance motor dives., and what must be checked before using it?
- What common mistake would make an answer or operation involving Magnet Motor drives, Switched reluctance motor dives. incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: Magnet Motor drives, Switched reluctance motor dives. [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 6: Identify various drive trains in Electric Vehicles and Hybrid Electric

Exact source point

> **Syllabus text:** Identify various drive trains in Electric Vehicles and Hybrid Electric

How to understand and study it

This point is an official syllabus requirement: “Identify various drive trains in Electric Vehicles and Hybrid Electric”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Identify various drive trains in Electric Vehicles, Hybrid Electric □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Identify various drive trains in Electric Vehicles and Hybrid Electric is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Identify various drive trains in Electric Vehicles and Hybrid Electric using the official terminology retained above.
- Organise the important elements of Identify various drive trains in Electric Vehicles and Hybrid Electric into a logical sequence, classification, structure, or calculation.
- Connect Identify various drive trains in Electric Vehicles and Hybrid Electric with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on Identify various drive trains in Electric Vehicles and Hybrid Electric with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Identify various drive trains in Electric Vehicles and Hybrid Electric. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Identify various drive trains in Electric Vehicles and Hybrid Electric is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Identify various drive trains in Electric Vehicles and Hybrid Electric, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Identify various drive trains in Electric Vehicles and Hybrid Electric, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Identify various drive trains in Electric Vehicles and Hybrid Electric?
- How would you distinguish Identify various drive trains in Electric Vehicles and Hybrid Electric from a closely related idea?
- Where would an engineer or technician encounter Identify various drive trains in Electric Vehicles and Hybrid Electric, and what must be checked before using it?
- What common mistake would make an answer or operation involving Identify various drive trains in Electric Vehicles and Hybrid Electric incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Identify various drive trains in Electric Vehicles and Hybrid Electric □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□□.

Learning goals

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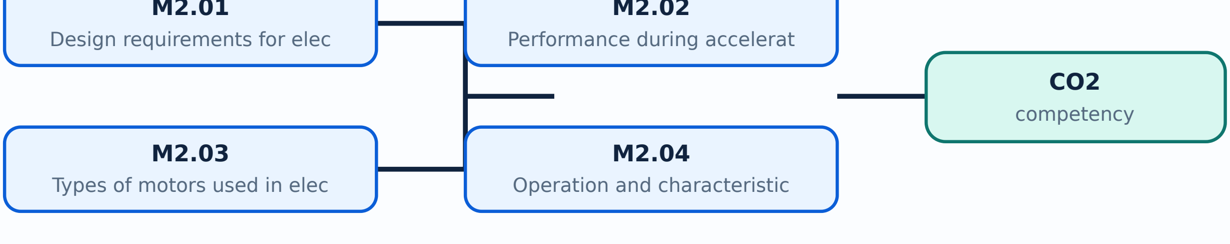
Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

M2.01 — Design requirements for electric vehicles (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Design requirements for electric vehicles in this topic. The required scope is: Range, maximum velocity, acceleration, power requirement, mass, resistances, tractive effort, transmission requirement, transmission efficiency and energy consumption.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Design requirements for electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Design requirements for electric vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□□.

Study points

- Define Design requirements for electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Design requirements for electric vehicles is applied in electric propulsion systems practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Design requirements for electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Design requirements for electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M2.01 — Design requirements for electric vehicles (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.01 — Design requirements for electric vehicles (3 hours) using the official terminology retained above.
- Organise the important elements of M2.01 — Design requirements for electric vehicles (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.01 — Design requirements for electric vehicles (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on M2.01 — Design requirements for electric vehicles (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.01 — Design requirements for electric vehicles (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.01 — Design requirements for electric vehicles (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.01 — Design requirements for electric vehicles (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.01 — Design requirements for electric vehicles (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.01 — Design requirements for electric vehicles (3 hours)?
- How would you distinguish M2.01 — Design requirements for electric vehicles (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.01 — Design requirements for electric vehicles (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.01 — Design requirements for electric vehicles (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.01 — Design requirements for electric vehicles (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours)

Concept and explanation

<p>Scope. The official syllabus places Performance during acceleration, coasting, moving up and down a hill in this topic. The required scope is: Performance of electric vehicles during acceleration, coasting, and motion up or down a hill. </p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Performance during acceleration, coasting, moving up and down a hill</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Performance during acceleration, coasting, moving up and down a hill ▯▯▯▯ topic ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ purpose, working principle, ▯▯▯▯▯▯ components ▯▯▯▯▯▯▯▯▯▯ variables, ▯▯▯▯▯▯ performance ▯▯▯▯▯▯▯▯▯▯ safety checks ▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯. Technical terms English-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯; diagram ▯▯▯▯▯▯▯▯▯▯ step sequence ▯▯▯▯▯▯▯▯▯▯ process ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯.

Study points

- Define Performance during acceleration, coasting, moving up and down a hill using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Performance during acceleration, coasting, moving up and down a hill is applied in electric propulsion systems practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Performance during acceleration, coasting, moving up and down a hill**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Performance during acceleration, coasting, moving up and down a hill without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) using the official terminology retained above.
- Organise the important elements of M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours)?
- How would you distinguish M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.02 — Performance during acceleration, coasting, moving up and down a hill (5 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M2.03 — Types of motors used in electric vehicles (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Types of motors used in electric vehicles in this topic. The required scope is: Classification of EV motors and selection considerations.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Types of motors used in electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Types of motors used in electric vehicles □□□□ topic □□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Types of motors used in electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Types of motors used in electric vehicles is applied in electric propulsion systems practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Types of motors used in electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Types of motors used in electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M2.03 — Types of motors used in electric vehicles (3 hours) is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M2.03 — Types of motors used in electric vehicles (3 hours) using the official terminology retained above.
- Organise the important elements of M2.03 — Types of motors used in electric vehicles (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.03 — Types of motors used in electric vehicles (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on M2.03 — Types of motors used in electric vehicles (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.03 — Types of motors used in electric vehicles (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M2.03 — Types of motors used in electric vehicles (3 hours) is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M2.03 — Types of motors used in electric vehicles (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.03 — Types of motors used in electric vehicles (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.03 — Types of motors used in electric vehicles (3 hours)?
- How would you distinguish M2.03 — Types of motors used in electric vehicles (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.03 — Types of motors used in electric vehicles (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.03 — Types of motors used in electric vehicles (3 hours) incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M2.03 — Types of motors used in electric vehicles (3 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□□ □□□□□□□□□□□□□.

M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours)

Concept and explanation

<p>Scope. The official syllabus places Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives in this topic. The required scope is: Configuration and control of DC motor drives, induction motor drives, permanent magnet motor drives, and switched reluctance motor drives.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr> <th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. </td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody> </table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.

- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives is applied in electric propulsion systems practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) using the official terminology retained above.
- Organise the important elements of M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) with one engineering decision, operation, measurement, or safety requirement in the context of EV Design and Motor Drives.
- Write an examination answer on M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours)?
- How would you distinguish M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.

- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M2.04 — Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives (5 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Module summary: Use tractive effort and motor characteristics to justify a drive choice.

OFFICIAL MODULE 3

Power Electronics, Drive Trains and Braking

Power converters, control systems, drive-train architecture, regenerative braking, and service practice connect the energy source to the road wheels.

Hours: 14
CO3 · Bloom level: Understanding

Handbook study routine: Complete Module 3 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 3

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

:
Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles.
Control systems for EV - Sizing the drive system - Sizing the propulsion motor, sizing the power electronics - Selecting the energy storage technology - Communications - supporting subsystems.
Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In
Wheel Drives.
The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid.
Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.
Maintenance, repairing, and services, Troubleshooting faults in different EV components
(Motor, drive train, battery, etc.).

Select the required energy storage and charging devices for Electric and

Point-by-point study guide

Official syllabus point 1: Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles.

Exact source point

Syllabus text: Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles.

How to understand and study it

This point is an official syllabus requirement: “Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. using the official terminology retained above.
- Organise the important elements of Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. into a logical sequence, classification, structure, or calculation.
- Connect Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles.. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles.?
- How would you distinguish Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. from a closely related idea?
- Where would an engineer or technician encounter Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles., and what must be checked before using it?
- What common mistake would make an answer or operation involving Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.

- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Power electronics related to EV - Rectifiers, Inverters, DC/DC converters, and Power Split devices for Hybrid Vehicles. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 2: Control systems for EV

Exact source point

Syllabus text: Control systems for EV

How to understand and study it

This point is an official syllabus requirement: “Control systems for EV”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Control systems for EV □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Control systems for EV should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

Learning objectives

- Define and scope Control systems for EV using the official terminology retained above.
- Organise the important elements of Control systems for EV into a logical sequence, classification, structure, or calculation.
- Connect Control systems for EV with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Control systems for EV with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Control systems for EV. It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

Engineering application lens

Engineering use of Control systems for EV depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

Worked answer method

Given: the quantity to be sensed, the operating environment, and the required decision or control action.
Required: a suitable measurement chain or an explanation of how the instrument operates.
Method: map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.
Answer: state the measured quantity, principle, important characteristics, application, and limitation.

Exam answer blueprint

For a short-answer question on Control systems for EV, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Control systems for EV, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Control systems for EV?
- How would you distinguish Control systems for EV from a closely related idea?
- Where would an engineer or technician encounter Control systems for EV, and what must be checked before using it?
- What common mistake would make an answer or operation involving Control systems for EV incorrect?

Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

Malayalam support: Control systems for EV [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 3: Sizing the drive system

Exact source point

Syllabus text: Sizing the drive system

How to understand and study it

This point is an official syllabus requirement: “Sizing the drive system”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ Sizing the drive system □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Sizing the drive system is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Sizing the drive system using the official terminology retained above.
- Organise the important elements of Sizing the drive system into a logical sequence, classification, structure, or calculation.
- Connect Sizing the drive system with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Sizing the drive system with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Sizing the drive system. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Sizing the drive system is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Sizing the drive system, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Sizing the drive system, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Sizing the drive system?
- How would you distinguish Sizing the drive system from a closely related idea?
- Where would an engineer or technician encounter Sizing the drive system, and what must be checked before using it?
- What common mistake would make an answer or operation involving Sizing the drive system incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Sizing the drive system □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 4: Sizing the propulsion motor, sizing the power electronics

Exact source point

Syllabus text: Sizing the propulsion motor, sizing the power electronics

How to understand and study it

This point is an official syllabus requirement: “Sizing the propulsion motor, sizing the power electronics”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Sizing the propulsion motor, sizing the power electronics □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Sizing the propulsion motor, sizing the power electronics must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Sizing the propulsion motor, sizing the power electronics using the official terminology retained above.
- Organise the important elements of Sizing the propulsion motor, sizing the power electronics into a logical sequence, classification, structure, or calculation.
- Connect Sizing the propulsion motor, sizing the power electronics with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Sizing the propulsion motor, sizing the power electronics with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Sizing the propulsion motor, sizing the power electronics. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Sizing the propulsion motor, sizing the power electronics is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Sizing the propulsion motor, sizing the power electronics, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Sizing the propulsion motor, sizing the power electronics, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Sizing the propulsion motor, sizing the power electronics?
- How would you distinguish Sizing the propulsion motor, sizing the power electronics from a closely related idea?
- Where would an engineer or technician encounter Sizing the propulsion motor, sizing the power electronics, and what must be checked before using it?
- What common mistake would make an answer or operation involving Sizing the propulsion motor, sizing the power electronics incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Sizing the propulsion motor, sizing the power electronics □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 5: Selecting the energy storage technology

Exact source point

Syllabus text: Selecting the energy storage technology

How to understand and study it

This point is an official syllabus requirement: “Selecting the energy storage technology”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Selecting the energy storage technology □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Selecting the energy storage technology must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Selecting the energy storage technology using the official terminology retained above.
- Organise the important elements of Selecting the energy storage technology into a logical sequence, classification, structure, or calculation.
- Connect Selecting the energy storage technology with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Selecting the energy storage technology with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Selecting the energy storage technology. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Selecting the energy storage technology is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Selecting the energy storage technology, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Selecting the energy storage technology, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Selecting the energy storage technology?
- How would you distinguish Selecting the energy storage technology from a closely related idea?
- Where would an engineer or technician encounter Selecting the energy storage technology, and what must be checked before using it?
- What common mistake would make an answer or operation involving Selecting the energy storage technology incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Selecting the energy storage technology □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 6: Communications

Exact source point

Syllabus text: Communications

How to understand and study it

This point is an official syllabus requirement: “Communications”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Communications □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Communications should be followed as a signal or information path. Explain what is generated, how it is modified or transmitted, what can disturb it, and how the receiver reconstructs or uses it.

Learning objectives

- Define and scope Communications using the official terminology retained above.
- Organise the important elements of Communications into a logical sequence, classification, structure, or calculation.
- Connect Communications with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Communications with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Communications. It is a study organisation method, not an additional official syllabus requirement.

- Define the information-bearing signal and the relevant source or channel.
- Describe the blocks or stages in order.
- Explain the role of bandwidth, noise, attenuation, distortion, coding, modulation, or protocol rules where applicable.
- Compare performance using range, capacity, reliability, complexity, cost, and application.

Engineering application lens

Engineering systems use Communications when information must be transferred reliably between equipment, instruments, controllers, or users. The choice of method is a balance between signal quality, distance, data rate, interference, infrastructure, safety, and maintenance.

Worked answer method

Given: source, channel, receiver, signal condition, or communication requirement.
Required: block diagram, operating explanation, comparison, or fault analysis.
Method: trace the signal from source to destination and mark where conversion, loss, interference, or recovery occurs.
Answer: state the principle, blocks, performance factors, application, and limitation.

Exam answer blueprint

For a short-answer question on Communications, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Communications, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Communications?
- How would you distinguish Communications from a closely related idea?
- Where would an engineer or technician encounter Communications, and what must be checked before using it?
- What common mistake would make an answer or operation involving Communications incorrect?

Common mistakes and safety emphasis

- Using transmitter, channel, and receiver as labels without explaining their functions.
- Confusing bandwidth, data rate, frequency, and range.
- Ignoring noise, attenuation, interference, synchronization, or protocol compatibility.
- Comparing systems without using common criteria.

Malayalam support: Communications [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 7: supporting subsystems.

Exact source point

Syllabus text: supporting subsystems.

How to understand and study it

This point is an official syllabus requirement: “supporting subsystems.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ supporting subsystems. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

supporting subsystems. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope supporting subsystems. using the official terminology retained above.
- Organise the important elements of supporting subsystems. into a logical sequence, classification, structure, or calculation.
- Connect supporting subsystems. with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on supporting subsystems. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading supporting subsystems.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of supporting subsystems. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on supporting subsystems., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is supporting subsystems., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining supporting subsystems.?
- How would you distinguish supporting subsystems. from a closely related idea?
- Where would an engineer or technician encounter supporting subsystems., and what must be checked before using it?
- What common mistake would make an answer or operation involving supporting subsystems. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: supporting subsystems. [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 8: Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration

Exact source point

Syllabus text: Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration

How to understand and study it

This point is an official syllabus requirement: “Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Drive train in EV, Basic Architecture of Electric Drive Trains, General configuration of an electric vehicle, Alternatives Based on Drive train Configuration □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration using the official terminology retained above.
- Organise the important elements of Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration into a logical sequence, classification, structure, or calculation.
- Connect Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on

Drive train Configuration is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration?
- How would you distinguish Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration from a closely related idea?
- Where would an engineer or technician encounter Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, and what must be checked before using it?

- What common mistake would make an answer or operation involving Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 9: Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In

Exact source point

Syllabus text: Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In

How to understand and study it

This point is an official syllabus requirement: “Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Alternatives Based on Power Source Configuration, Single, Multi-motor Drives - In □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In using the official terminology retained above.
- Organise the important elements of Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In into a logical sequence, classification, structure, or calculation.
- Connect Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In?
- How would you distinguish Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In from a closely related idea?
- Where would an engineer or technician encounter Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In, and what must be checked before using it?
- What common mistake would make an answer or operation involving Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 10: Wheel Drives.

Exact source point

Syllabus text: Wheel Drives.

How to understand and study it

This point is an official syllabus requirement: “Wheel Drives.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Wheel Drives. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Wheel Drives. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Wheel Drives. using the official terminology retained above.
- Organise the important elements of Wheel Drives. into a logical sequence, classification, structure, or calculation.
- Connect Wheel Drives. with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Wheel Drives. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Wheel Drives.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Wheel Drives. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Wheel Drives., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Wheel Drives., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Wheel Drives.?
- How would you distinguish Wheel Drives. from a closely related idea?
- Where would an engineer or technician encounter Wheel Drives., and what must be checked before using it?
- What common mistake would make an answer or operation involving Wheel Drives. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Wheel Drives. □□□□□ topic □□□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□□ □□□□□□–□□□□ □□□□□□ □□□□□□□□□□□□□□□.

Official syllabus point 11: The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid.

Exact source point

Syllabus text: The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid.

How to understand and study it

This point is an official syllabus requirement: “The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid.”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. using the official terminology retained above.
- Organise the important elements of The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. into a logical sequence, classification, structure, or calculation.
- Connect The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid.?
- How would you distinguish The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. from a closely related idea?
- Where would an engineer or technician encounter The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid., and what must be checked before using it?
- What common mistake would make an answer or operation involving The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.

- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: The architecture of hybrid vehicle drive train – Series hybrid, Parallel hybrid, Series-Parallel hybrid, and Complex hybrid. □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□– □□□ □□□□□ □□□□□□□□□□□□□□□.

Official syllabus point 12: Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.

Exact source point

Syllabus text: Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.

How to understand and study it

This point is an official syllabus requirement: “Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Braking system in electric, Hybrid vehicles, Regenerative braking, Hybrid braking. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. using the official terminology retained above.
- Organise the important elements of Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. into a logical sequence, classification, structure, or calculation.
- Connect Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.?
- How would you distinguish Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. from a closely related idea?
- Where would an engineer or technician encounter Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking., and what must be checked before using it?
- What common mistake would make an answer or operation involving Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 13: Maintenance, repairing, and services, Troubleshooting faults in different EV components

Exact source point

> **Syllabus text:** Maintenance, repairing, and services, Troubleshooting faults in different EV components

How to understand and study it

This point is an official syllabus requirement: “Maintenance, repairing, and services, Troubleshooting faults in different EV components”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Maintenance, repairing, and services, Troubleshooting faults in different EV components □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Maintenance, repairing, and services, Troubleshooting faults in different EV components must be understood as a control-of-risk topic. Identify the hazard, the possible harm, the conditions that increase the risk, and the hierarchy of controls that prevents the incident.

Learning objectives

- Define and scope Maintenance, repairing, and services, Troubleshooting faults in different EV components using the official terminology retained above.
- Organise the important elements of Maintenance, repairing, and services, Troubleshooting faults in different EV components into a logical sequence, classification, structure, or calculation.
- Connect Maintenance, repairing, and services, Troubleshooting faults in different EV components with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Maintenance, repairing, and services, Troubleshooting faults in different EV components with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Maintenance, repairing, and services, Troubleshooting faults in different EV components. It is a study organisation method, not an additional official syllabus requirement.

- Identify the source of energy or unsafe condition.
- Describe the possible consequence and the people, equipment, or environment exposed.
- Apply elimination, substitution, engineering control, administrative control, and personal protective equipment in that order.
- State inspection, isolation, emergency response, reporting, and recovery requirements where relevant.

Engineering application lens

In engineering practice, Maintenance, repairing, and services, Troubleshooting faults in different EV components is part of normal design, operation, maintenance, and troubleshooting—not a separate final paragraph. The safest answer connects the control measure to the hazard and explains why the measure is effective.

Worked answer method

Given: the work activity, equipment, hazard, or incident condition.
Required: identify risk and propose controls.
Method: describe hazard → consequence → control → verification → emergency action.
Answer: give specific controls, responsible action, and one condition that must be checked before work begins.

Exam answer blueprint

For a short-answer question on Maintenance, repairing, and services, Troubleshooting faults in different EV components, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Maintenance, repairing, and services, Troubleshooting faults in different EV components, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Maintenance, repairing, and services, Troubleshooting faults in different EV components?
- How would you distinguish Maintenance, repairing, and services, Troubleshooting faults in different EV components from a closely related idea?
- Where would an engineer or technician encounter Maintenance, repairing, and services, Troubleshooting faults in different EV components, and what must be checked before using it?
- What common mistake would make an answer or operation involving Maintenance, repairing, and services, Troubleshooting faults in different EV components incorrect?

Common mistakes and safety emphasis

- Naming PPE as the only control when isolation or guarding is required.
- Describing a hazard without stating the consequence.
- Ignoring stored energy, restart, access, housekeeping, or emergency isolation.
- Treating a safety instruction as optional or disconnected from the operating procedure.

Malayalam support: Maintenance, repairing, and services, Troubleshooting faults in different EV components □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 14: (Motor, drive train, battery, etc.).

Exact source point

Syllabus text: (Motor, drive train, battery, etc.).

How to understand and study it

This point is an official syllabus requirement: “(Motor, drive train, battery, etc.).”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ (Motor, drive train, battery, etc.). □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

(Motor, drive train, battery, etc.). is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope (Motor, drive train, battery, etc.). using the official terminology retained above.
- Organise the important elements of (Motor, drive train, battery, etc.). into a logical sequence, classification, structure, or calculation.
- Connect (Motor, drive train, battery, etc.). with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on (Motor, drive train, battery, etc.). with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading (Motor, drive train, battery, etc.).. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, (Motor, drive train, battery, etc.). is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.
Required: construction, working, selection, performance, or diagnosis as requested.
Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.
Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on (Motor, drive train, battery, etc.)., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is (Motor, drive train, battery, etc.)., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining (Motor, drive train, battery, etc.).?
- How would you distinguish (Motor, drive train, battery, etc.). from a closely related idea?
- Where would an engineer or technician encounter (Motor, drive train, battery, etc.)., and what must be checked before using it?
- What common mistake would make an answer or operation involving (Motor, drive train, battery, etc.). incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: (Motor, drive train, battery, etc.). □□□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□□ definition □□□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□□. Exam answer □□□□□□□□□□□□ concept-□□□□□ □□□□□□–□□□□ □□□□□□ □□□□□□□□□□□□□□□□.

Official syllabus point 15: Select the required energy storage and charging devices for Electric and

Exact source point

> **Syllabus text:** Select the required energy storage and charging devices for Electric and

How to understand and study it

This point is an official syllabus requirement: “Select the required energy storage and charging devices for Electric and”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□ Select the required energy storage, charging devices for Electric and □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Select the required energy storage and charging devices for Electric and must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Select the required energy storage and charging devices for Electric and using the official terminology retained above.
- Organise the important elements of Select the required energy storage and charging devices for Electric and into a logical sequence, classification, structure, or calculation.
- Connect Select the required energy storage and charging devices for Electric and with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on Select the required energy storage and charging devices for Electric and with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Select the required energy storage and charging devices for Electric and. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Select the required energy storage and charging devices for Electric and is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Select the required energy storage and charging devices for Electric and, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Select the required energy storage and charging devices for Electric and, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Select the required energy storage and charging devices for Electric and?
- How would you distinguish Select the required energy storage and charging devices for Electric and from a closely related idea?
- Where would an engineer or technician encounter Select the required energy storage and charging devices for Electric and, and what must be checked before using it?
- What common mistake would make an answer or operation involving Select the required energy storage and charging devices for Electric and incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Select the required energy storage and charging devices for Electric and [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Learning goals

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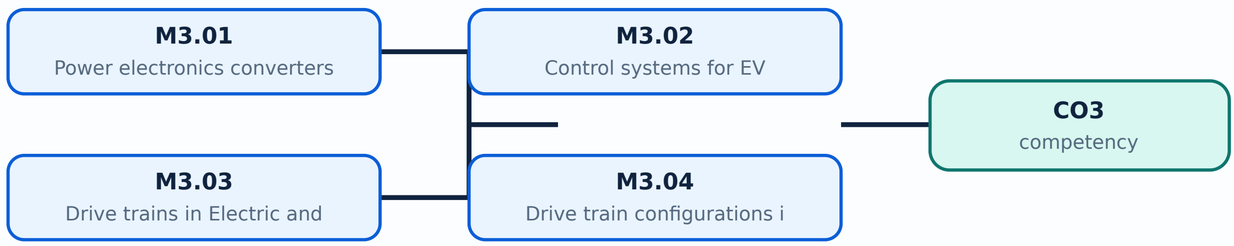
Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Power electronics converters used in electric and hybrid vehicles in this topic. The required scope is: Rectifiers, inverters, DC/DC converters, and power-split devices for hybrid vehicles.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Power electronics converters used in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Power electronics converters used in electric and hybrid vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Power electronics converters used in electric and hybrid vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Power electronics converters used in electric and hybrid vehicles is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Power electronics converters used in electric and hybrid vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Power electronics converters used in electric and hybrid vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) using the official terminology retained above.
- Organise the important elements of M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours)?
- How would you distinguish M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M3.01 — Power electronics converters used in electric and hybrid vehicles (2 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

M3.02 — Control systems for EV (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Control systems for EV in this topic. The required scope is: Sizing the drive system, propulsion motor, power electronics, energy-storage technology, communications, and supporting subsystems. </p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Control systems for EV</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. </td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody> </table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Control systems for EV □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Control systems for EV using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Control systems for EV is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Control systems for EV**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Control systems for EV without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M3.02 — Control systems for EV (2 hours) should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

Learning objectives

- Define and scope M3.02 — Control systems for EV (2 hours) using the official terminology retained above.
- Organise the important elements of M3.02 — Control systems for EV (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.02 — Control systems for EV (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on M3.02 — Control systems for EV (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.02 — Control systems for EV (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

Engineering application lens

Engineering use of M3.02 — Control systems for EV (2 hours) depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

Worked answer method

Given: the quantity to be sensed, the operating environment, and the required decision or control action.
Required: a suitable measurement chain or an explanation of how the instrument operates.
Method: map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.
Answer: state the measured quantity, principle, important characteristics, application, and limitation.

Exam answer blueprint

For a short-answer question on M3.02 — Control systems for EV (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.02 — Control systems for EV (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.02 — Control systems for EV (2 hours)?
- How would you distinguish M3.02 — Control systems for EV (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.02 — Control systems for EV (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.02 — Control systems for EV (2 hours) incorrect?

Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

Malayalam support: M3.02 — Control systems for EV (2 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Drive trains in Electric and Hybrid Electric Vehicles in this topic. The required scope is: Basic architecture, general EV configuration, alternatives based on drive-train and power-source configuration, single/multi-motor and in-wheel drives.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"> <table><caption>Study frame for Drive trains in Electric and Hybrid Electric Vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th> <td>How the input is converted, controlled, transported, stored, measured, or protected. </td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr> <th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p> Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Drive trains in Electric and Hybrid Electric Vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Drive trains in Electric and Hybrid Electric Vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Drive trains in Electric and Hybrid Electric Vehicles is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Drive trains in Electric and Hybrid Electric Vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Drive trains in Electric and Hybrid Electric Vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) using the official terminology retained above.
- Organise the important elements of M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours)?
- How would you distinguish M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.03 — Drive trains in Electric and Hybrid Electric Vehicles (2 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Drive train configurations in electric and hybrid electric vehicles in this topic. The required scope is: Series, parallel, series-parallel, and complex hybrid drive trains.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Drive train configurations in electric and hybrid electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Drive train configurations in electric and hybrid electric vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Drive train configurations in electric and hybrid electric vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Drive train configurations in electric and hybrid electric vehicles is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Drive train configurations in electric and hybrid electric vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Drive train configurations in electric and hybrid electric vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) using the official terminology retained above.
- Organise the important elements of M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours)?
- How would you distinguish M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.04 — Drive train configurations in electric and hybrid electric vehicles (3 hours) ▯▯▯▯ topic ▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ exact technical term ▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯ principle, named parts/steps, application, limitation ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯. English terminology, symbols, units, diagram labels ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. Exam answer ▯▯▯▯▯▯▯▯▯▯ concept-▯▯▯▯ ▯▯▯▯▯–▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯.

M3.05 — Regenerative braking and hybrid braking (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Regenerative braking and hybrid braking in this topic. The required scope is: Braking in electric and hybrid vehicles, energy recovery and coordination with friction braking.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Regenerative braking and hybrid braking</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Regenerative braking and hybrid braking □□□□ topic □□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Regenerative braking and hybrid braking using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Regenerative braking and hybrid braking is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Regenerative braking and hybrid braking**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Regenerative braking and hybrid braking without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M3.05 — Regenerative braking and hybrid braking (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.05 — Regenerative braking and hybrid braking (2 hours) using the official terminology retained above.
- Organise the important elements of M3.05 — Regenerative braking and hybrid braking (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.05 — Regenerative braking and hybrid braking (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on M3.05 — Regenerative braking and hybrid braking (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.05 — Regenerative braking and hybrid braking (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.05 — Regenerative braking and hybrid braking (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.05 — Regenerative braking and hybrid braking (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.05 — Regenerative braking and hybrid braking (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.05 — Regenerative braking and hybrid braking (2 hours)?
- How would you distinguish M3.05 — Regenerative braking and hybrid braking (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.05 — Regenerative braking and hybrid braking (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.05 — Regenerative braking and hybrid braking (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.05 — Regenerative braking and hybrid braking (2 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M3.06 — Service and maintenance of EV components (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Service and maintenance of EV components in this topic. The required scope is: Maintenance, repair, service, and troubleshooting of motor, drive train, battery, and other EV components.</p> <p> Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Service and maintenance of EV components</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr> <th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked. </td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody> </table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Malayalam support: Malayalam support: Service and maintenance of EV components □□□□ topic □□□□□□□□□□□□ □□□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□□ variables, □□□□□□□ performance □□□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□; diagram □□□□□□□□□□□ step sequence □□□□□□□□□□□ process □□□□□□□□□□□□ □□□□□□□□□□□□□□□.

Study points

- Define Service and maintenance of EV components using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Service and maintenance of EV components is applied in ev powertrain engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Service and maintenance of EV components**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Service and maintenance of EV components without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M3.06 — Service and maintenance of EV components (3 hours) must be understood as a control-of-risk topic. Identify the hazard, the possible harm, the conditions that increase the risk, and the hierarchy of controls that prevents the incident.

Learning objectives

- Define and scope M3.06 — Service and maintenance of EV components (3 hours) using the official terminology retained above.
- Organise the important elements of M3.06 — Service and maintenance of EV components (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.06 — Service and maintenance of EV components (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Power Electronics, Drive Trains and Braking.
- Write an examination answer on M3.06 — Service and maintenance of EV components (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.06 — Service and maintenance of EV components (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the source of energy or unsafe condition.
- Describe the possible consequence and the people, equipment, or environment exposed.
- Apply elimination, substitution, engineering control, administrative control, and personal protective equipment in that order.
- State inspection, isolation, emergency response, reporting, and recovery requirements where relevant.

Engineering application lens

In engineering practice, M3.06 — Service and maintenance of EV components (3 hours) is part of normal design, operation, maintenance, and troubleshooting—not a separate final paragraph. The safest answer connects the control measure to the hazard and explains why the measure is effective.

Worked answer method

Given: the work activity, equipment, hazard, or incident condition.
Required: identify risk and propose controls.
Method: describe hazard → consequence → control → verification → emergency action.
Answer: give specific controls, responsible action, and one condition that must be checked before work begins.

Exam answer blueprint

For a short-answer question on M3.06 — Service and maintenance of EV components (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.06 — Service and maintenance of EV components (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.06 — Service and maintenance of EV components (3 hours)?
- How would you distinguish M3.06 — Service and maintenance of EV components (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.06 — Service and maintenance of EV components (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.06 — Service and maintenance of EV components (3 hours) incorrect?

Common mistakes and safety emphasis

- Naming PPE as the only control when isolation or guarding is required.
- Describing a hazard without stating the consequence.
- Ignoring stored energy, restart, access, housekeeping, or emergency isolation.
- Treating a safety instruction as optional or disconnected from the operating procedure.

Malayalam support: M3.06 — Service and maintenance of EV components (3 hours) □□□□ topic □□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□.

Module summary: Trace power flow from battery through converter and motor to wheels, including regenerative return.

OFFICIAL MODULE 4

Energy Storage, Fuel Cells and Charging

Energy storage determines range, power capability, charging time, safety, cost, and lifecycle of an EV or HEV.

Hours: 16
CO4 · Bloom level: Understanding

Handbook study routine: Complete Module 4 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 4

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

:
Energy storage requirements in electric and hybrid vehicles. Definition of various Battery
Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors –
Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells.

Battery Management System, Battery charging technologies: Methods of battery charging –
Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station,

Point-by-point study guide

Official syllabus point 1: Energy storage requirements in electric and hybrid vehicles. Definition of various Battery

Exact source point

> **Syllabus text:** Energy storage requirements in electric and hybrid vehicles. Definition of various Battery

How to understand and study it

This point is an official syllabus requirement: “Energy storage requirements in electric and hybrid vehicles. Definition of various Battery”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Energy storage requirements in electric, hybrid vehicles. Definition of various Battery □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Energy storage requirements in electric and hybrid vehicles. Definition of various Battery must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Energy storage requirements in electric and hybrid vehicles. Definition of various Battery using the official terminology retained above.
- Organise the important elements of Energy storage requirements in electric and hybrid vehicles. Definition of various Battery into a logical sequence, classification, structure, or calculation.
- Connect Energy storage requirements in electric and hybrid vehicles. Definition of various Battery with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on Energy storage requirements in electric and hybrid vehicles. Definition of various Battery with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Energy storage requirements in electric and hybrid vehicles. Definition of various Battery. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Energy storage requirements in electric and hybrid vehicles. Definition of various Battery is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Energy storage requirements in electric and hybrid vehicles. Definition of various Battery, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Energy storage requirements in electric and hybrid vehicles. Definition of various Battery, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Energy storage requirements in electric and hybrid vehicles. Definition of various Battery?
- How would you distinguish Energy storage requirements in electric and hybrid vehicles. Definition of various Battery from a closely related idea?
- Where would an engineer or technician encounter Energy storage requirements in electric and hybrid vehicles. Definition of various Battery, and what must be checked before using it?
- What common mistake would make an answer or operation involving Energy storage requirements in electric and hybrid vehicles. Definition of various Battery incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Energy storage requirements in electric and hybrid vehicles. Definition of various Battery [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 2: Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors –

Exact source point

Syllabus text: Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors –

How to understand and study it

This point is an official syllabus requirement: "Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors –". Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – should be studied by linking structure, property, process, and application. The important question is why a material or process gives a particular behaviour and where that behaviour is useful or limiting.

Learning objectives

- Define and scope Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – using the official terminology retained above.
- Organise the important elements of Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – into a logical sequence, classification, structure, or calculation.
- Connect Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors –. It is a study organisation method, not an additional official syllabus requirement.

- Identify the material, process, property, or defect named in the topic.
- Explain the relevant structure or mechanism at the level expected in the syllabus.
- Relate the mechanism to strength, hardness, conductivity, corrosion, manufacturability, durability, or another stated property.
- Finish with selection criteria, application, limitation, and safety or quality consideration.

Engineering application lens

In engineering practice, Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – affects selection, manufacturing quality, service life, inspection, and maintenance. A technically useful answer links the observed property or defect to its cause and then to the method used to control or exploit it.

Worked answer method

Given: the material, process, property, or service condition in the question.
Required: explanation, comparison, selection, or identification of the relevant mechanism.
Method: state the principle; connect cause to property; compare alternatives using the same criteria; mention the relevant test or control.
Answer: give the conclusion with the application and the principal limitation.

Exam answer blueprint

For a short-answer question on Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors –, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors –, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors –?
- How would you distinguish Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – from a closely related idea?
- Where would an engineer or technician encounter Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal

Air based. Battery charging, Quick Charging devices. Ultra-capacitors –, and what must be checked before using it?

- What common mistake would make an answer or operation involving Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – incorrect?

Common mistakes and safety emphasis

- Memorising a property without its unit, condition, or engineering meaning.
- Confusing a manufacturing process with a material property.
- Giving a comparison with different criteria for different materials.
- Ignoring defects, surface condition, heat treatment, environment, or quality control.

Malayalam support: Parameters. Different types of batteries – Lead Acid, Nickel-based, Sodium based, Lithium-based, Metal Air based. Battery charging, Quick Charging devices. Ultra-capacitors – □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 3: Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells.

Exact source point

> **Syllabus text:** Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells.

How to understand and study it

This point is an official syllabus requirement: “Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells.”. Study the sequence from input or initial condition to operation and final output. Note the role of each named stage and the cause-and-effect relationship. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□□ Features, basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle, operation, Types of Fuel Cells. □□□□ technical terms English-□□□□□□□□ □□□□□□□□. □□□□□□ definition □□□□□□□□□□□ principle □□□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Diagram, sequence, formula, unit □□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□ revision □□□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. using the official terminology retained above.
- Organise the important elements of Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. into a logical sequence, classification, structure, or calculation.
- Connect Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells.. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells.?
- How would you distinguish Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. from a closely related idea?
- Where would an engineer or technician encounter Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of

different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells., and what must be checked before using it?

- What common mistake would make an answer or operation involving Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: Features and basic principle. Ultrahigh-speed flywheel-based energy storage. Hybridization of different energy Storages. Fuel Cells- basic principle and operation, Types of Fuel Cells. □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 4: Battery Management System, Battery charging technologies: Methods of battery charging –

Exact source point

> **Syllabus text:** Battery Management System, Battery charging technologies: Methods of battery charging –

How to understand and study it

This point is an official syllabus requirement: “Battery Management System, Battery charging technologies: Methods of battery charging –”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Battery Management System, Battery charging technologies, Methods of battery charging – □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Battery Management System, Battery charging technologies: Methods of battery charging – is a decision and coordination topic. Learn the definitions, then connect the planning inputs, method, output, performance measure, and corrective action.

Learning objectives

- Define and scope Battery Management System, Battery charging technologies: Methods of battery charging – using the official terminology retained above.
- Organise the important elements of Battery Management System, Battery charging technologies: Methods of battery charging – into a logical sequence, classification, structure, or calculation.
- Connect Battery Management System, Battery charging technologies: Methods of battery charging – with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on Battery Management System, Battery charging technologies: Methods of battery charging – with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Battery Management System, Battery charging technologies: Methods of battery charging –. It is a study organisation method, not an additional official syllabus requirement.

- Define the management term or objective.
- Identify the resources, constraints, people, information, and time involved.
- Describe the procedure or decision criteria in a logical sequence.
- Measure the result and state how deviations are controlled or improved.

Engineering application lens

Engineering organisations apply Battery Management System, Battery charging technologies: Methods of battery charging – to deliver safe, economical, reliable, and timely work. A strong answer connects the method to productivity, quality, cost, schedule, customer requirement, compliance, or sustainability as appropriate.

Worked answer method

Given: the project, process, resource, or organisational situation.
Required: plan, comparison, decision, or performance explanation.
Method: state objective → inputs → method → output → measure → corrective action.
Answer: finish with the likely benefit, limitation, and condition for successful implementation.

Exam answer blueprint

For a short-answer question on Battery Management System, Battery charging technologies: Methods of battery charging –, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Battery Management System, Battery charging technologies: Methods of battery charging –, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Battery Management System, Battery charging technologies: Methods of battery charging –?
- How would you distinguish Battery Management System, Battery charging technologies: Methods of battery charging – from a closely related idea?
- Where would an engineer or technician encounter Battery Management System, Battery charging technologies: Methods of battery charging –, and what must be checked before using it?
- What common mistake would make an answer or operation involving Battery Management System, Battery charging technologies: Methods of battery charging – incorrect?

Common mistakes and safety emphasis

- Listing management terms without a sequence or decision criterion.
- Ignoring constraints, measurement, responsibility, or feedback.
- Confusing an objective with an activity or a result with an input.
- Giving a generic benefit without linking it to the engineering situation.

Malayalam support: Battery Management System, Battery charging technologies: Methods of battery charging – [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 5: Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station

Exact source point

Syllabus text: Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station

How to understand and study it

This point is an official syllabus requirement: “Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ Onboard, offboard charging, Domestic charging infrastructure, Public charging infrastructure □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station should be studied by linking structure, property, process, and application. The important question is why a material or process gives a particular behaviour and where that behaviour is useful or limiting.

Learning objectives

- Define and scope Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station using the official terminology retained above.
- Organise the important elements of Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station into a logical sequence, classification, structure, or calculation.
- Connect Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station. It is a study organisation method, not an additional official syllabus requirement.

- Identify the material, process, property, or defect named in the topic.
- Explain the relevant structure or mechanism at the level expected in the syllabus.
- Relate the mechanism to strength, hardness, conductivity, corrosion, manufacturability, durability, or another stated property.
- Finish with selection criteria, application, limitation, and safety or quality consideration.

Engineering application lens

In engineering practice, Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station affects selection, manufacturing quality, service life, inspection, and maintenance. A technically useful answer links the observed property or defect to its cause and then to the method used to control or exploit it.

Worked answer method

Given: the material, process, property, or service condition in the question.
Required: explanation, comparison, selection, or identification of the relevant mechanism.
Method: state the principle; connect cause to property; compare alternatives using the same criteria; mention the relevant test or control.
Answer: give the conclusion with the application and the principal limitation.

Exam answer blueprint

For a short-answer question on Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station?
- How would you distinguish Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station from a closely related idea?
- Where would an engineer or technician encounter Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station, and what must be checked before using it?
- What common mistake would make an answer or operation involving Onboard and offboard charging, Domestic charging infrastructure, Public

charging infrastructure, Fast charging station, Battery swapping station incorrect?

Common mistakes and safety emphasis

- Memorising a property without its unit, condition, or engineering meaning.
- Confusing a manufacturing process with a material property.
- Giving a comparison with different criteria for different materials.
- Ignoring defects, surface condition, heat treatment, environment, or quality control.

Malayalam support: Onboard and offboard charging, Domestic charging infrastructure, Public charging infrastructure, Fast charging station, Battery swapping station □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Learning goals

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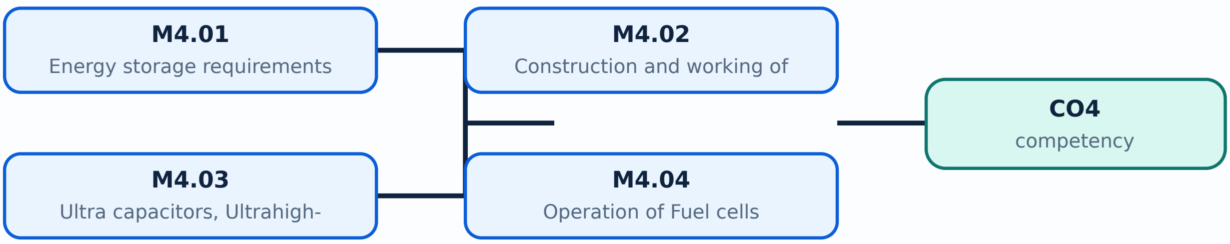
Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Energy storage requirements in electric and hybrid vehicles in this topic. The required scope is: Energy-storage requirements and battery parameters for EV and HEV applications.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Energy storage requirements in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Malayalam support: Malayalam support: Energy storage requirements in electric and hybrid vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Energy storage requirements in electric and hybrid vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Energy storage requirements in electric and hybrid vehicles is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Energy storage requirements in electric and hybrid vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Energy storage requirements in electric and hybrid vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) using the official terminology retained above.
- Organise the important elements of M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.

Required: the unknown quantity, including its unit.

Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.

Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours)?
- How would you distinguish M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M4.01 — Energy storage requirements in electric and hybrid vehicles (2 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours)

Concept and explanation

<p>Scope. The official syllabus places Construction and working of batteries used in electric and hybrid vehicles in this topic. The required scope is: Lead-acid, nickel-based, sodium-based, lithium-based, and metal-air batteries; construction, operation, and characteristics.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Construction and working of batteries used in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr> <tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Construction and working of batteries used in electric and hybrid vehicles □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□□.

Study points

- Define Construction and working of batteries used in electric and hybrid vehicles using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Construction and working of batteries used in electric and hybrid vehicles is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Construction and working of batteries used in electric and hybrid vehicles**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Construction and working of batteries used in electric and hybrid vehicles without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) using the official terminology retained above.
- Organise the important elements of M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours)?
- How would you distinguish M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.02 — Construction and working of batteries used in electric and hybrid vehicles (5 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours)

Concept and explanation

<p>Scope. The official syllabus places Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages in this topic. The required scope is: Features and principle of ultracapacitors, flywheel storage, and hybridization of storage devices.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr> <tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□□.

Study points

- Define Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) using the official terminology retained above.
- Organise the important elements of M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours)?
- How would you distinguish M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M4.03 — Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M4.04 — Operation of Fuel cells (2 hours)

Concept and explanation

<p>Scope. The official syllabus places Operation of Fuel cells in this topic. The required scope is: Basic principle, operation, and types of fuel cells.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Operation of Fuel cells</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr> <tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p> Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Operation of Fuel cells □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Operation of Fuel cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Operation of Fuel cells is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Operation of Fuel cells**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Operation of Fuel cells without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M4.04 — Operation of Fuel cells (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.04 — Operation of Fuel cells (2 hours) using the official terminology retained above.
- Organise the important elements of M4.04 — Operation of Fuel cells (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.04 — Operation of Fuel cells (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on M4.04 — Operation of Fuel cells (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.04 — Operation of Fuel cells (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.04 — Operation of Fuel cells (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.04 — Operation of Fuel cells (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.04 — Operation of Fuel cells (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.04 — Operation of Fuel cells (2 hours)?
- How would you distinguish M4.04 — Operation of Fuel cells (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.04 — Operation of Fuel cells (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.04 — Operation of Fuel cells (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.04 — Operation of Fuel cells (2 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

− M4.05 — Battery charging Technologies and Battery management systems (4 hours)

Concept and explanation

<p>Scope. The official syllabus places Battery charging Technologies and Battery management systems in this topic. The required scope is: Onboard/offboard charging, domestic/public infrastructure, fast charging, battery swapping, and battery-management systems.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"> <table><caption>Study frame for Battery charging Technologies and Battery management systems</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr> <tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

> **Malayalam support:** Malayalam support: Battery charging Technologies and Battery management systems □□□□ topic □□□□□□□□□□□□ □□□□□ □□□□□□□ purpose, working principle, □□□□□□ components □□□□□□□□□□ variables, □□□□□□ performance □□□□□□□□□□ safety checks □□□□ □□□□□□□□□□ □□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□; diagram □□□□□□□□□□ step sequence □□□□□□□□□□ process □□□□□□□□□□□ □□□□□□□□□□□□□.

Study points

- Define Battery charging Technologies and Battery management systems using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Battery charging Technologies and Battery management systems is applied in ev energy storage practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Battery charging Technologies and Battery management systems**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Battery charging Technologies and Battery management systems without explaining the working sequence, variables, or engineering reason for each item.

Handbook treatment

M4.05 — Battery charging Technologies and Battery management systems (4 hours) is a decision and coordination topic. Learn the definitions, then connect the planning inputs, method, output, performance measure, and corrective action.

Learning objectives

- Define and scope M4.05 — Battery charging Technologies and Battery management systems (4 hours) using the official terminology retained above.
- Organise the important elements of M4.05 — Battery charging Technologies and Battery management systems (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.05 — Battery charging Technologies and Battery management systems (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Energy Storage, Fuel Cells and Charging.
- Write an examination answer on M4.05 — Battery charging Technologies and Battery management systems (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.05 — Battery charging Technologies and Battery management systems (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the management term or objective.
- Identify the resources, constraints, people, information, and time involved.
- Describe the procedure or decision criteria in a logical sequence.
- Measure the result and state how deviations are controlled or improved.

Engineering application lens

Engineering organisations apply M4.05 — Battery charging Technologies and Battery management systems (4 hours) to deliver safe, economical, reliable, and timely work. A strong answer connects the method to productivity, quality, cost, schedule, customer requirement, compliance, or sustainability as appropriate.

Worked answer method

Given: the project, process, resource, or organisational situation.
Required: plan, comparison, decision, or performance explanation.
Method: state objective → inputs → method → output → measure → corrective action.
Answer: finish with the likely benefit, limitation, and condition for successful implementation.

Exam answer blueprint

For a short-answer question on M4.05 — Battery charging Technologies and Battery management systems (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.05 — Battery charging Technologies and Battery management systems (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.05 — Battery charging Technologies and Battery management systems (4 hours)?
- How would you distinguish M4.05 — Battery charging Technologies and Battery management systems (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.05 — Battery charging Technologies and Battery management systems (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.05 — Battery charging Technologies and Battery management systems (4 hours) incorrect?

Common mistakes and safety emphasis

- Listing management terms without a sequence or decision criterion.
- Ignoring constraints, measurement, responsibility, or feedback.
- Confusing an objective with an activity or a result with an input.
- Giving a generic benefit without linking it to the engineering situation.

Malayalam support: M4.05 — Battery charging Technologies and Battery management systems (4 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Module summary: Select storage and charging technology against range, power, safety, and infrastructure constraints.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

'ule to view its labelled learning-map diagram.

Module 2: EV Design and

rning-map diagram.

Module 3: Power

its labelled learning-map diagram.

Module 4: Energy Storage,

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list for this theory course. Use the topic procedures, worked examples, diagrams, and module questions in this lesson for structured study.

Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment for this theory course.

Assessment Methodology

The supplied PDF does not specify a separate CIE/ESE distribution.

- The supplied PDF lists Series Test – I and Series Test – II entries, but a complete official CIE/ESE mark distribution and question-paper pattern are not specified in the supplied PDF.
- The practice model paper in this lesson is therefore explicitly a study aid, not an official examination paper.

Module-wise Questions and Answers

module-1 — Electric and Hybrid Vehicle Fundamentals

1. Explain Need for hybrid and electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Need for hybrid and electric vehicles in this topic. The required scope is: Need for hybrid and electric vehicles; historical background; social and environmental importance.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition. </p> <div class="table-wrap"><table><caption>Study frame for Need for hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr> </thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th>

<td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Components and working principle of electric and hybrid electric vehicles. [3 marks]

<p>Scope. The official syllabus places Components and working principle of electric and hybrid electric vehicles in this topic. The required scope is: Main components and working principles of electric and hybrid vehicles.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Components and working principle of electric and hybrid electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Configurations of electric vehicles and hybrid vehicles. [3 marks]

<p>Scope. The official syllabus places Configurations of electric vehicles and hybrid vehicles in this topic. The required scope is: Pure Electric vehicle (PE), Battery Electric vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid Electric vehicle (HEV), Plug-in Hybrid Electric vehicle (PHEV), and types based on differential; series, parallel, series-parallel and complex HEV.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Configurations of electric vehicles and hybrid vehicles</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Compare petrol, diesel, hybrid and electric vehicles. [3 marks]

<p>Scope. The official syllabus places Compare petrol, diesel, hybrid and electric vehicles in this topic. The required scope is: Comparative study of EV and conventional vehicles; differences between complete EV and hybrid vehicles; life cycle assessment.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system

named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Compare petrol, diesel, hybrid and electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric vehicle engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — EV Design and Motor Drives

5. Explain Design requirements for electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Design requirements for electric vehicles in this topic. The required scope is: Range, maximum velocity, acceleration, power requirement, mass, resistances, tractive effort, transmission requirement, transmission efficiency and energy consumption.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Design requirements for electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component,

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Performance during acceleration, coasting, moving up and down a hill. **[3 marks]**

<p>Scope. The official syllabus places Performance during acceleration, coasting, moving up and down a hill in this topic. The required scope is: Performance of electric vehicles during acceleration, coasting, and motion up or down a hill. </p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Performance during acceleration, coasting, moving up and down a hill</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Types of motors used in electric vehicles. **[3 marks]**

<p>Scope. The official syllabus places Types of motors used in electric vehicles in this topic. The required scope is: Classification of EV motors and selection considerations.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Types of motors used in electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in electric propulsion systems.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives. **[3 marks]**

<p>Scope. The official syllabus places Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives in this topic. The required scope is: Configuration and control of DC motor drives, induction motor drives, permanent magnet motor drives, and switched reluctance motor drives.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives</caption><thead><tr><th>Aspect</th><th>What

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Power Electronics, Drive Trains and Braking

9. Explain Power electronics converters used in electric and hybrid vehicles. [3 marks]

<p>Scope. The official syllabus places Power electronics converters used in electric and hybrid vehicles in this topic. The required scope is: Rectifiers, inverters, DC/DC converters, and power-split devices for hybrid vehicles.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Power electronics converters used in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship

first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Control systems for EV. **[3 marks]**

<p>Scope. The official syllabus places Control systems for EV in this topic. The required scope is: Sizing the drive system, propulsion motor, power electronics, energy-storage technology, communications, and supporting subsystems. </p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Control systems for EV</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Drive trains in Electric and Hybrid Electric Vehicles. **[3 marks]**

<p>Scope. The official syllabus places Drive trains in Electric and Hybrid Electric Vehicles in this topic. The required scope is: Basic architecture, general EV configuration, alternatives based on drive-train and power-source configuration, single/multi-motor and in-wheel drives.</p> <p>Concept and method.

Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Drive trains in Electric and Hybrid Electric Vehicles</caption> <thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr> <th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Drive train configurations in electric and hybrid electric vehicles. [3 marks]

<p>Scope. The official syllabus places Drive train configurations in electric and hybrid electric vehicles in this topic. The required scope is: Series, parallel, series-parallel, and complex hybrid drive trains.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Drive train configurations in electric and hybrid electric vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev powertrain engineering.</td></tr><tr> <th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Energy Storage, Fuel Cells and Charging

13. Explain Energy storage requirements in electric and hybrid vehicles. [3 marks]

<p>Scope. The official syllabus places Energy storage requirements in electric and hybrid vehicles in this topic. The required scope is: Energy-storage requirements and battery parameters for EV and HEV applications.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Energy storage requirements in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Construction and working of batteries used in electric and hybrid vehicles. [3 marks]

<p>Scope. The official syllabus places Construction and working of batteries used in electric and hybrid vehicles in this topic. The required scope is: Lead-acid, nickel-based, sodium-based, lithium-based, and metal-air batteries; construction, operation, and characteristics.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table> <caption>Study frame for Construction and working of batteries used in electric and hybrid vehicles</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead> <tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr> <th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages. [3 marks]

<p>Scope. The official syllabus places Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages in this topic. The required scope is: Features and principle of ultracapacitors, flywheel storage, and hybridization of storage devices.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit,

assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Operation of Fuel cells. [3 marks]

<p>Scope. The official syllabus places Operation of Fuel cells in this topic. The required scope is: Basic principle, operation, and types of fuel cells.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Operation of Fuel cells</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in ev energy storage.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Need for hybrid and electric vehicles* with one relevant application. (5 marks)
2. **2.** Define or explain *Components and working principle of electric and hybrid electric vehicles* with one relevant application. (5 marks)
3. **3.** Define or explain *Design requirements for electric vehicles* with one relevant application. (5 marks)
4. **4.** Define or explain *Performance during acceleration, coasting, moving up and down a hill* with one relevant application. (5 marks)
5. **5.** Define or explain *Power electronics converters used in electric and hybrid vehicles* with one relevant application. (5 marks)
6. **6.** Define or explain *Control systems for EV* with one relevant application. (5 marks)
7. **7.** Define or explain *Energy storage requirements in electric and hybrid vehicles* with one relevant application. (5 marks)
8. **8.** Define or explain *Construction and working of batteries used in electric and hybrid vehicles* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Electric and Hybrid Vehicle Fundamentals:** Electric vehicle architecture and policy comparison.

- **EV Design and Motor Drives:** Use tractive effort and motor characteristics to justify a drive choice.
- **Power Electronics, Drive Trains and Braking:** Trace power flow from battery through converter and motor to wheels, including regenerative return.
- **Energy Storage, Fuel Cells and Charging:** Select storage and charging technology against range, power, safety, and infrastructure constraints.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Need for hybrid and electric vehicles	<p>Scope. The official syllabus places Need for hybrid and electric vehicles in this topic. The required scope is: Need for hybrid and electric vehicles; historical background; social and environmental importance.</p> <p>Conce
Components and working principle of electric and hybrid electric vehicles	<p>Scope. The official syllabus places Components and working principle of electric and hybrid electric vehicles in this topic. The required scope is: Main components and working principles of electric and hybrid vehicles.</p> <p><str
Configurations of electric vehicles and hybrid vehicles	<p>Scope. The official syllabus places Configurations of electric vehicles and hybrid vehicles in this topic. The required scope is: Pure Electric vehicle (PE), Battery Electric vehicle (BEV), Fuel Cell Electric Vehicle (FCEV), Hybrid
Compare petrol, diesel, hybrid and electric vehicles	<p>Scope. The official syllabus places Compare petrol, diesel, hybrid and electric vehicles in this topic. The required scope is: Comparative study of EV and conventional vehicles; differences between complete EV and hybrid vehicles;

Term	Short meaning
Advantages and disadvantages of hybrid and electric vehicles	<p>Scope. The official syllabus places Advantages and disadvantages of hybrid and electric vehicles in this topic. The required scope is: Technical, economic, environmental, range, charging, maintenance, and lifecycle advantages and l
Government policies and schemes related to Electric vehicles	<p>Scope. The official syllabus places Government policies and schemes related to Electric vehicles in this topic. The required scope is: FAME-1, FAME-2, Transformative Mobility and Energy Storage Mission, central and state policies,
Design requirements for electric vehicles	<p>Scope. The official syllabus places Design requirements for electric vehicles in this topic. The required scope is: Range, maximum velocity, acceleration, power requirement, mass, resistances, tractive effort, transmission requirem
Performance during acceleration, coasting, moving up and down a hill	<p>Scope. The official syllabus places Performance during acceleration, coasting, moving up and down a hill in this topic. The required scope is: Performance of electric vehicles during acceleration, coasting, and motion up or down a
Types of motors used in electric vehicles	<p>Scope. The official syllabus places Types of motors used in electric vehicles in this topic. The required scope is: Classification of EV motors and selection considerations.</p> <p> Concept and method. Study this to
Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives	<p>Scope. The official syllabus places Operation and characteristics of DC, Induction, Permanent Magnet and Switched Reluctance motor drives in this topic. The required scope is: Configuration and control of DC motor drives, induction
Power electronics converters used in electric and hybrid vehicles	Scope. The official syllabus places Power electronics converters used in electric and hybrid vehicles in this topic. The required scope is: Rectifiers, inverters, DC/DC converters, and power-split devices for hybrid vehicles. <
Control systems for EV	<p>Scope. The official syllabus places Control systems for EV in this topic. The required scope is: Sizing the drive system, propulsion motor, power electronics, energy-storage technology, communications, and supporting subsystems.</p
Drive trains in Electric and Hybrid Electric Vehicles	<p>Scope. The official syllabus places Drive trains in Electric and Hybrid Electric Vehicles in this topic. The required scope is: Basic architecture, general EV configuration, alternatives based on drive-train and power-source config
Drive train configurations in electric and hybrid electric vehicles	<p>Scope. The official syllabus places Drive train configurations in electric and hybrid electric vehicles in this topic. The required scope is: Series, parallel, series-parallel, and complex hybrid drive trains.</p> <p>Concep
Regenerative braking and hybrid braking	<p>Scope. The official syllabus places Regenerative braking and hybrid braking in this topic. The

Term	Short meaning
	required scope is: Braking in electric and hybrid vehicles, energy recovery and coordination with friction braking. Conce
Service and maintenance of EV components	<p>Scope. The official syllabus places Service and maintenance of EV components in this topic. The required scope is: Maintenance, repair, service, and troubleshooting of motor, drive train, battery, and other EV components.</p> <p><s
Energy storage requirements in electric and hybrid vehicles	<p>Scope. The official syllabus places Energy storage requirements in electric and hybrid vehicles in this topic. The required scope is: Energy-storage requirements and battery parameters for EV and HEV applications.</p> <p>Co
Construction and working of batteries used in electric and hybrid vehicles	<p>Scope. The official syllabus places Construction and working of batteries used in electric and hybrid vehicles in this topic. The required scope is: Lead-acid, nickel-based, sodium-based, lithium-based, and metal-air batteries; con
Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages	<p>Scope. The official syllabus places Ultra capacitors, Ultrahigh-speed Flywheel and Hybridization of energy storages in this topic. The required scope is: Features and principle of ultracapacitors, flywheel storage, and hybridizatio
Operation of Fuel cells	<p>Scope. The official syllabus places Operation of Fuel cells in this topic. The required scope is: Basic principle, operation, and types of fuel cells.</p> <p>Concept and method. Study this topic as a sequence of <s
Battery charging Technologies and Battery management systems	<p>Scope. The official syllabus places Battery charging Technologies and Battery management systems in this topic. The required scope is: Onboard/offboard charging, domestic/public infrastructure, fast charging, battery swapping, and

Official References and Resources

References listed in the supplied syllabus

1. Jack Erjavec and Jeff Arias, Hybrid, Electric and Fuel-cell Vehicles, Cengage Learning India Pvt Ltd.
2. C. Mi, M. A. Masrur and D. W. Gao, Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives, Wiley, 2011.
3. S. Onori, L. Serrao and G. Rizzoni, Hybrid Electric Vehicles: Energy Management Strategies, Springer, 2015.
4. M. Ehsani et al., Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, CRC Press, 2004.
5. T. Denton, Electric and Hybrid Vehicles, Routledge.
6. Iqbal Hussein, Electric and Hybrid Vehicles Design Fundamentals, CRC Press.

7. A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, 2018.

Official learning resources listed

- https://nptel.ac.in/courses/108/103/108103009/
- https://nptel.ac.in/courses/108/106/108106170

In-page Search